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Closed-Loop Neurofeedback System

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Feedback: The engine of learning

Feedback happens when the **result of an action flows back to shape the next action**. It is how we learn almost everything without instructions.

A baby learning to walk

- No lecture on the theory of walking.
- Tries, wobbles, and senses body position.
- Adjusts and improves step by step.

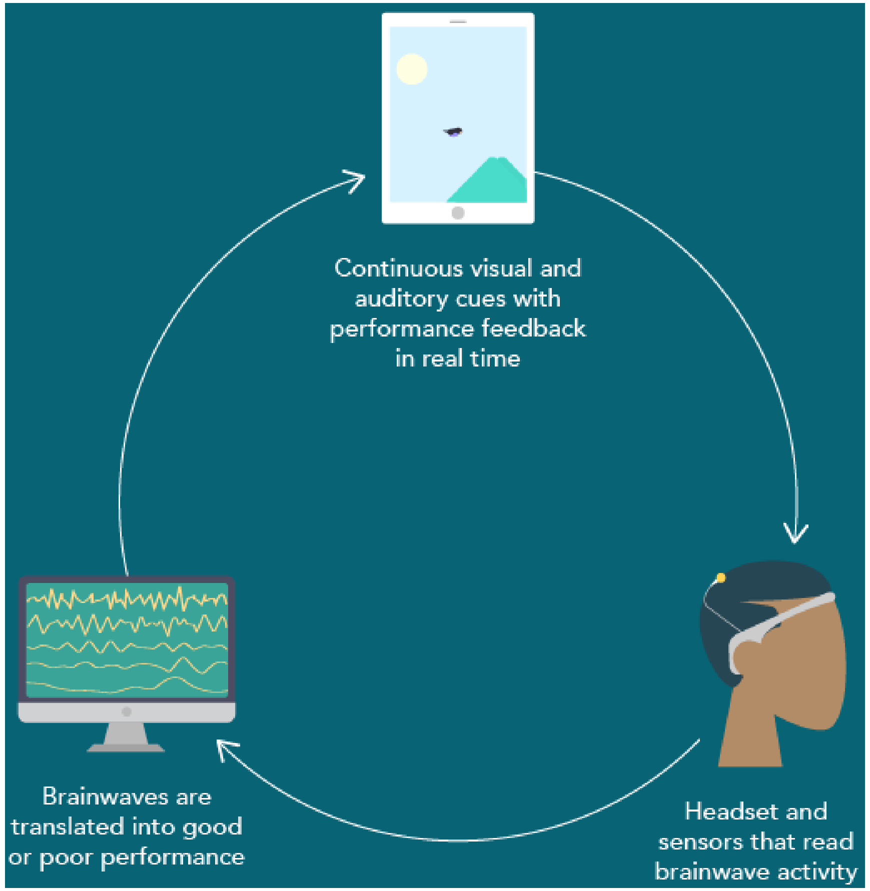
The key point

- Learning needs a signal to **steer by**.
- The faster and clearer the feedback, the faster the learner improves.

Biofeedback: Measure, see, and change

Biofeedback is feedback about a **body signal**.

A device measures something **you cannot normally sense**, shows it to you, and that awareness lets you change it.



The biofeedback family

Electrocardiogram (ECG) / Heart Rate Variability (HRV)	• Heart rate and its variability
Electromyography (EMG)	• Muscle tension
Electrodermal activity (EDA)	• Skin conductance (sweat)
Temperature	• Peripheral blood flow
Hemoencephalography (HEG)	• Brain blood flow
Electroencephalogram (EEG)	• Brain electrical activity

What neurofeedback is, and is not?

Neurofeedback is **EEG biofeedback** on the brain's own electrical rhythms.

It measures	It does not inject	It is not a 'treatment'
• Electrodes record tiny scalp voltages. • Software tracks a target rhythm live. • The signal is shown back as feedback.	• No current or stimulation enters the brain. • The brain simply sees its own activity. • This is why it is inherently gentle.	• The device does not fix anything. • It gives the brain information; the brain does the changing.

Arousal and the need for flexibility

Slow waves calm us; fast waves energize us. Problems come from being stuck too much of either, at the wrong time.

The goal of neurofeedback is **promoting the right rhythm for the right circumstance**.

Delta	Theta	Alpha	Beta	Gamma

◀ **slower waves · calm, drowsy, internal**

faster waves · alert, engaged, external ▶

Too much slow wave

When alert: Trouble concentrating and finishing tasks.

Widespread theta: The brain isn't 'switching on', focus drifts.

Too much fast wave

When resting: Hard to calm down or fall asleep.

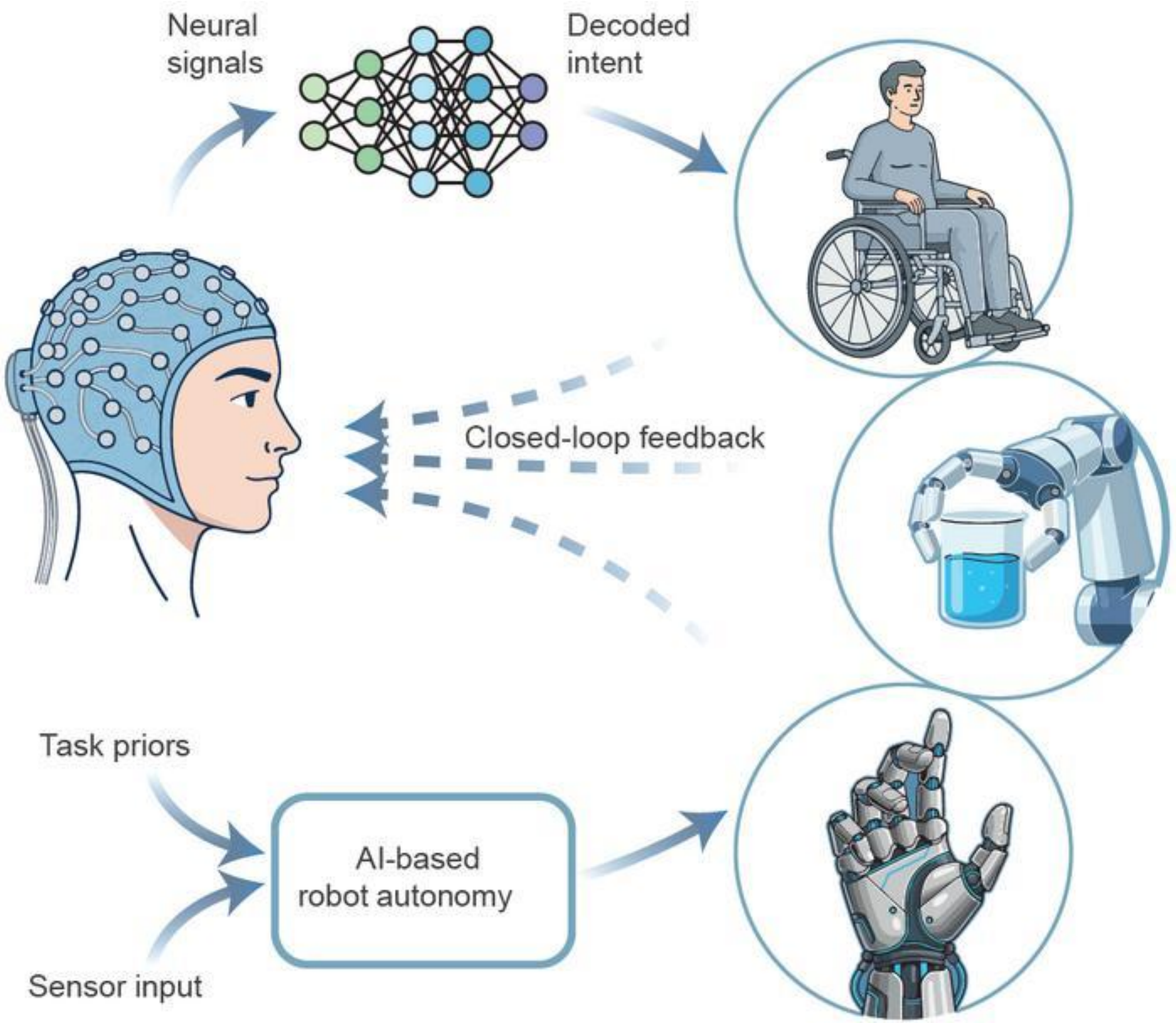
Excess gamma: Agitation, anxiety, and even aggression.

The closed loop in real time

Many times, a second the system reads the brain, checks the target rhythm, and rewards the brain, the instant it moves the right way.

Nobody tells the brain how to do it – the **reward** alone steers it, just like the baby's first steps.

Delegating local kinematics to the robot's AI **minimizes the user's cognitive load**, letting them focus purely on their overarching goal.

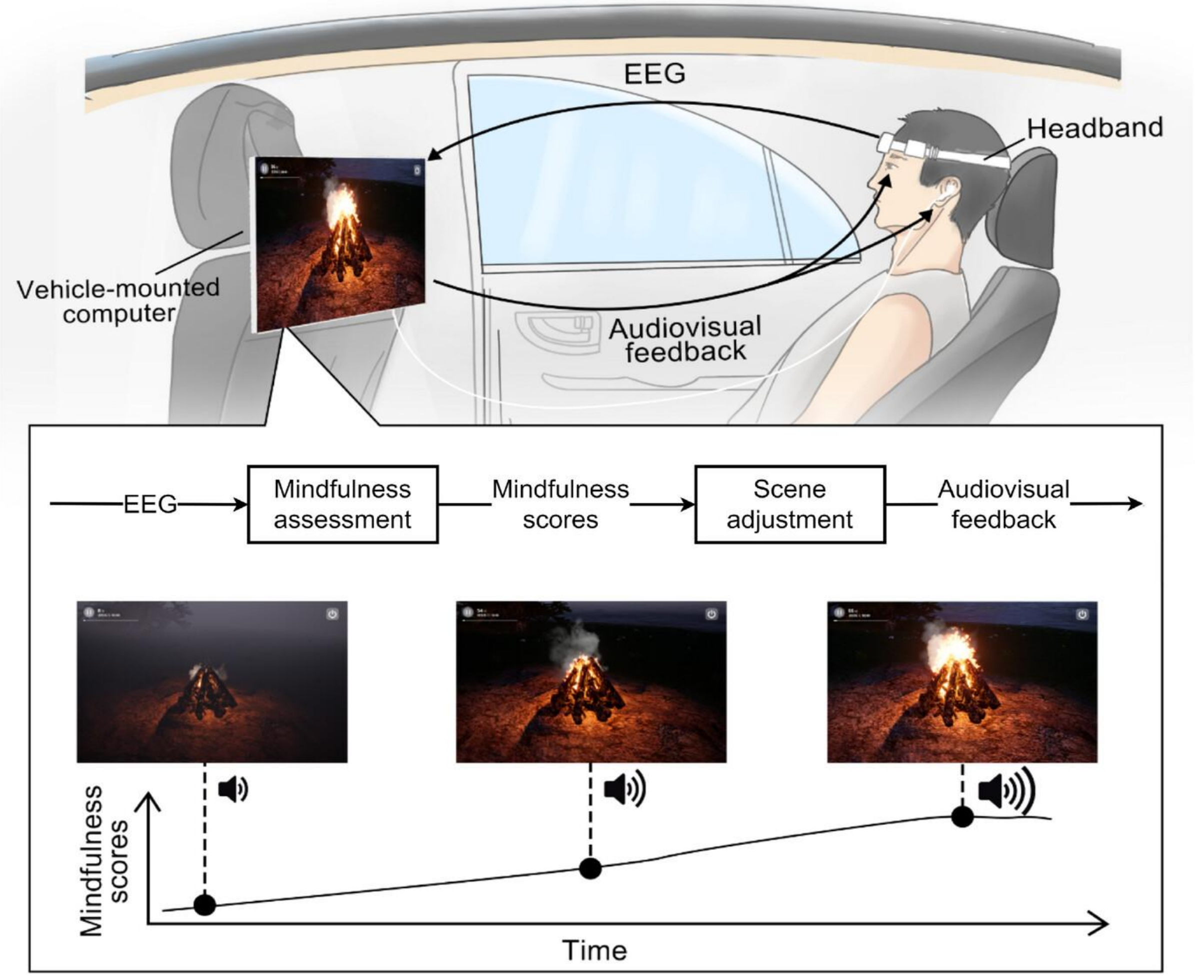


Rewarding a brain state

The loop is **reinforcement learning for the brain**. When a desired state appears and is instantly rewarded, the brain produces more of it.

The learning is mostly implicit – the person doesn’t consciously ‘try’; **the brain is shaped by the reward**, like learning to ride a bike.

You don’t force the state – you allow it and let the feedback confirm when you’ve found it.



Reward, thresholds, and motivation

Setting the threshold

Reward triggers when the target is met.	Aim for frequent but earned success.

Feedback that engages

A game advances, a video brightens, music plays, a bar fills, points score.

Keeping motivation

Too hard → Frustration and no learning.	Too easy → No signal to improve.	The sweet spot sustains attention.

What a neurofeedback training session looks like?

A session is a **guided workout** for the brain; **assess, set a target, train with reps, and review**.

Hands-on practitioners **tailor and adjust the protocol**; while plug-and-play systems decide automatically (**faster to learn and less individualized**).

Choose a protocol

- Pick the site and rhythm to encourage or discourage.

Connect sensors

- Place electrodes on the scalp; check signal quality.

Train with feedback

- Many short reps; the brain earns rewards for the target state.

Review and adjust

- Track change across sessions and fine-tune the plan.

Types of neurofeedback

Most methods share the same closed loop; they differ in **what they target and how much the practitioner steers**.

Amplitude (classical)

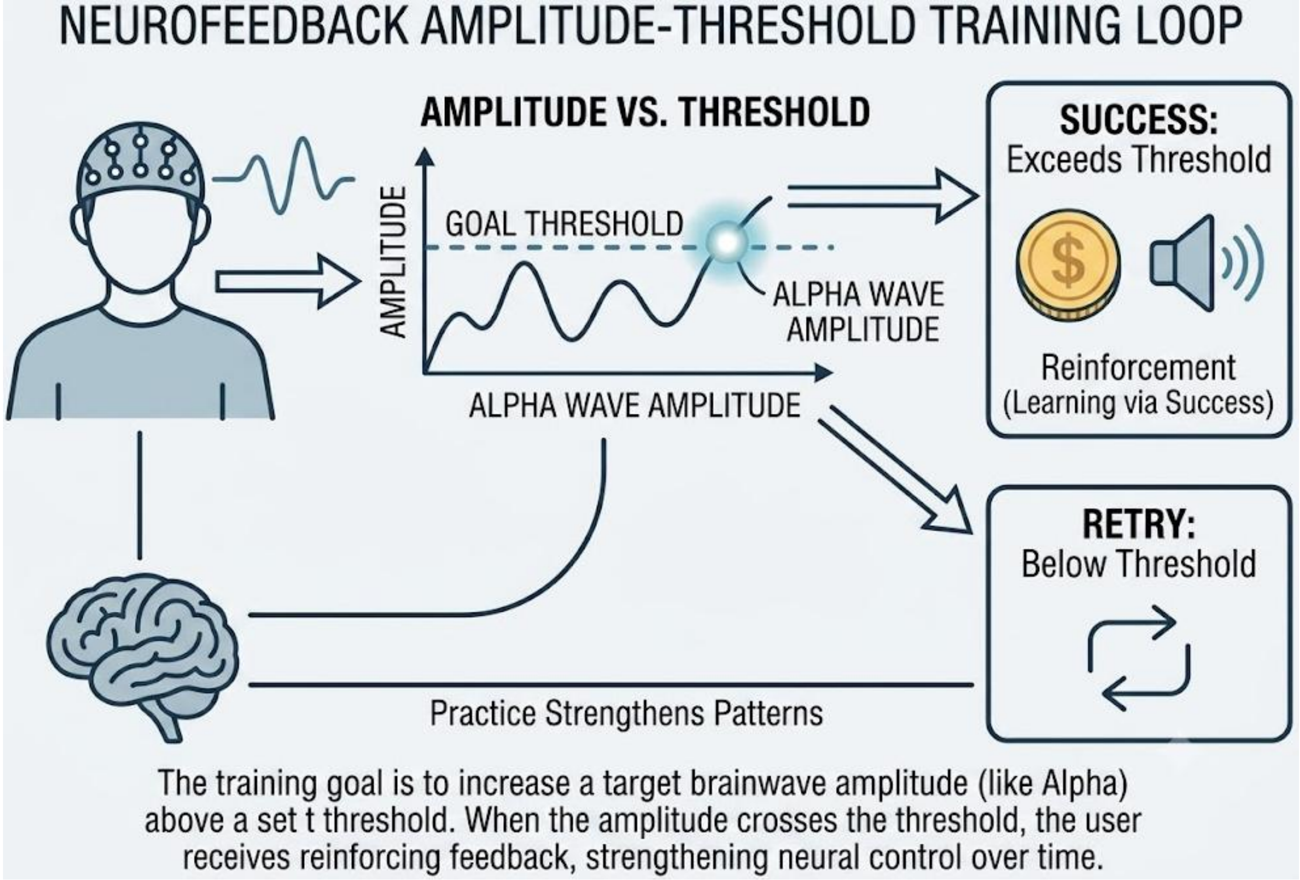
- Teaches users to **control brainwave rhythms**: Alpha for relaxation, Beta for focus.

SCP

- Trains the brain to **shift its baseline energy**, becoming more alert or rested.

fMRI / HEG

- Measures blood flow and oxygen changes to **show the hardest-working brain regions**.



Is neurofeedback safe?

‘Safe’ means the chance of a serious **unwanted effect is very low**. Because nothing is put into the brain, that risk is small, but task design still matters.

The method is gentle; the practitioner’s assessment and adjustment are what keep it both safe and effective.

Non-invasive

- Equipment only measures and displays.
- No surgery, no substances, and no current.

The gym analogy

- Untrained, you can strain yourself.
- With a good coach choosing the exercises, the risk drops and progress is safer.

Honest caveats

- Mild and temporary effects can occur, while serious effects are very rare.
- Safety depends on **how it is used**.

Key takeaways

The through-line: a fast, clear signal plus reward lets a plastic brain retune itself.

A closed loop	• Measure the brain, show it, and it can change it.
The brain does the work	• Feedback guides; neuroplasticity delivers the result.
Reward-driven learning	• Operant conditioning shapes states, mostly implicitly.
Right rhythm, right moment	• The goal is flexibility, not one ‘ideal’ wave.
Safe but skill-dependent	• Gentle by design; outcomes hinge on good practice.

Muse 2 online pipeline

Initial live test

System architecture and utilized libraries

os, sys, time	Core system operations and millisecond-precision timers.
multiprocessing, threading	Asynchronous hardware streaming and non-blocking execution.
collections (deque)	Fast, thread-safe rolling buffers for the decision logic.
numpy	High-performance matrix operations and mathematical aggregations.
scipy (signal)	Causal linear filtering (lfilter) and PSD extraction (welch).
pylsl, muselsl	Headset BLE handshakes and LSL continuous data acquisition.
joblib	Deserialization of the frozen BCI production bundle.
pygame, tkinter	Pre-registration UI and the low-latency, hardware-accelerated visual feedback engine.

The .joblib production bundle

The trained classifier

- The core **machine learning algorithm** (e.g., LDA) responsible for analyzing the processed signals and **calculating the real-time probability** of the target brain state.

The baseline scaler

- The **saved offline calibration metrics (mean and variance)**. It is packaged with the model to ensure **live brainwaves are consistently normalized against the user's specific resting baseline**.

The feature map

- The **strict, ordered list of required signal features** (e.g., specific Beta/Alpha channel ratios). It guarantees the **live data array is structured perfectly to match the exact matrix the model learned during training**.

Online evaluation pipeline overview

Unified task routing

- A single standalone engine dynamically routes participants to either the **complex mental math** or **missing letter (Cloze)** evaluation task based on the operator's initial selection.

Pre-flight signal checks

- A mandatory hardware validation phase that requires **5 continuous seconds of clean**, artifact-free data before the protocol can begin.

Dynamic visual UI

- A real-time dashboard that provides participants with **direct visual feedback** tied to the gears of the 4-step decision engine, aiding in neurofeedback learning.

Automated data persistence

- System metrics such as **ITR, end-to-end latency, and artifact rejection rates** are automatically generated and saved directly to the subject's directory as text reports.

Initialization and acquisition

Operator registration GUI

- A Tkinter interface captures the subject ID and the desired cognitive task, establishing the correct directory pathways.

Dynamic bundle loading

- The engine automatically searches for and loads the specific machine learning bundle corresponding to the selected task (e.g., *bci_live_production_math_lda_online.joblib* vs. the words equivalent).

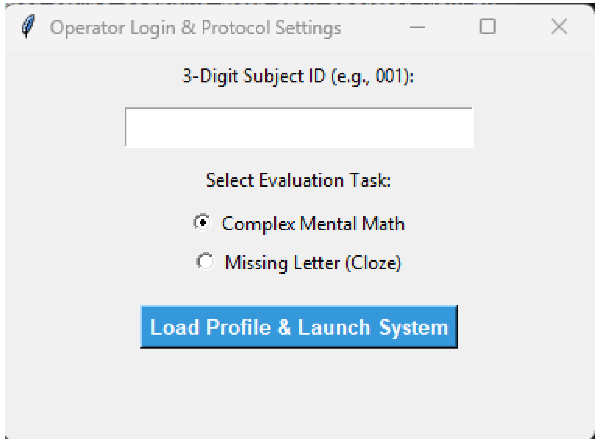
LSL stream resolution

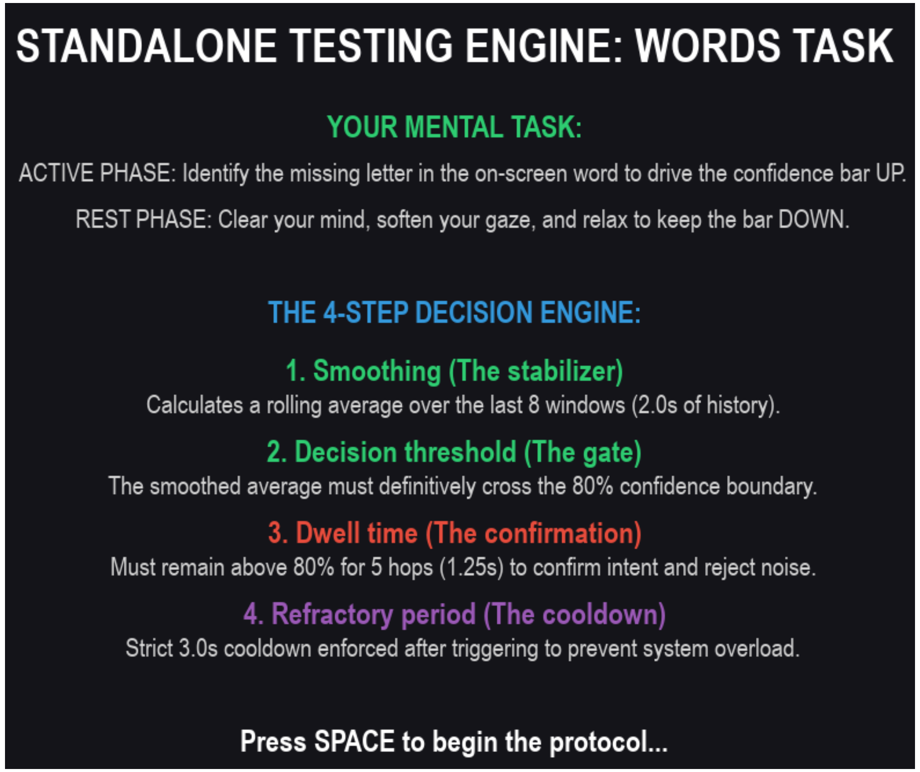
- Actively scans for and resolves the Muse headset's EEG stream over Bluetooth.

Hardware calibration phase

- Evaluates TP9, AF7, AF8, and TP10 in real-time, blocking the protocol's start until any flatlines (loose sensors) or high-voltage artifacts are physically corrected by the participant.

Initialization and acquisition





Real-time artifact refusal

Immediate amplitude detection

- LSL pulls raw EEG data in **0.25s hops**, maintaining a **2.0s rolling window**. The system checks if peak-to-peak variance exceeds the 350μV baseline limit.

Fail-safe command withholding

- If an **artifact is detected**, the window is **marked bad, dropped,** and the **machine learning prediction is completely skipped**.

Logic reset

- The UI warns the user, and the internal probability buffers are instantly flushed to **ensure an artifact spike cannot accidentally trigger a command**.

Processing and inference

Causal adherence

- Clean epochs pass through **strict causal filter** implementations (notch and bandpass) to prevent data look-ahead crashes.

Feature engineering

- Normalized Alpha and Beta PSD are extracted via Welch, ratio is calculated, and then integrated in the system.

Scaled probability output

- This **normalizes incoming brainwaves against the user's unique resting state**, ensuring the model reacts to genuine relative neural shifts rather than absolute background voltage.
- Forces the model to output a **precise decimal probability** (e.g., 0.85 for 85% certainty) of the "**Active Focus**" state, replacing rigid binary guesses with absolute mathematical confidence.

The decision engine

Smoothing (The stabilizer)

- **Raw probabilities** enter a deque to calculate **a <u>rolling average over the last 8 windows</u> (2.0 seconds of history)**.

Decision threshold (The gate)

- The **smoothed average** must definitively cross the **80%** confidence boundary.

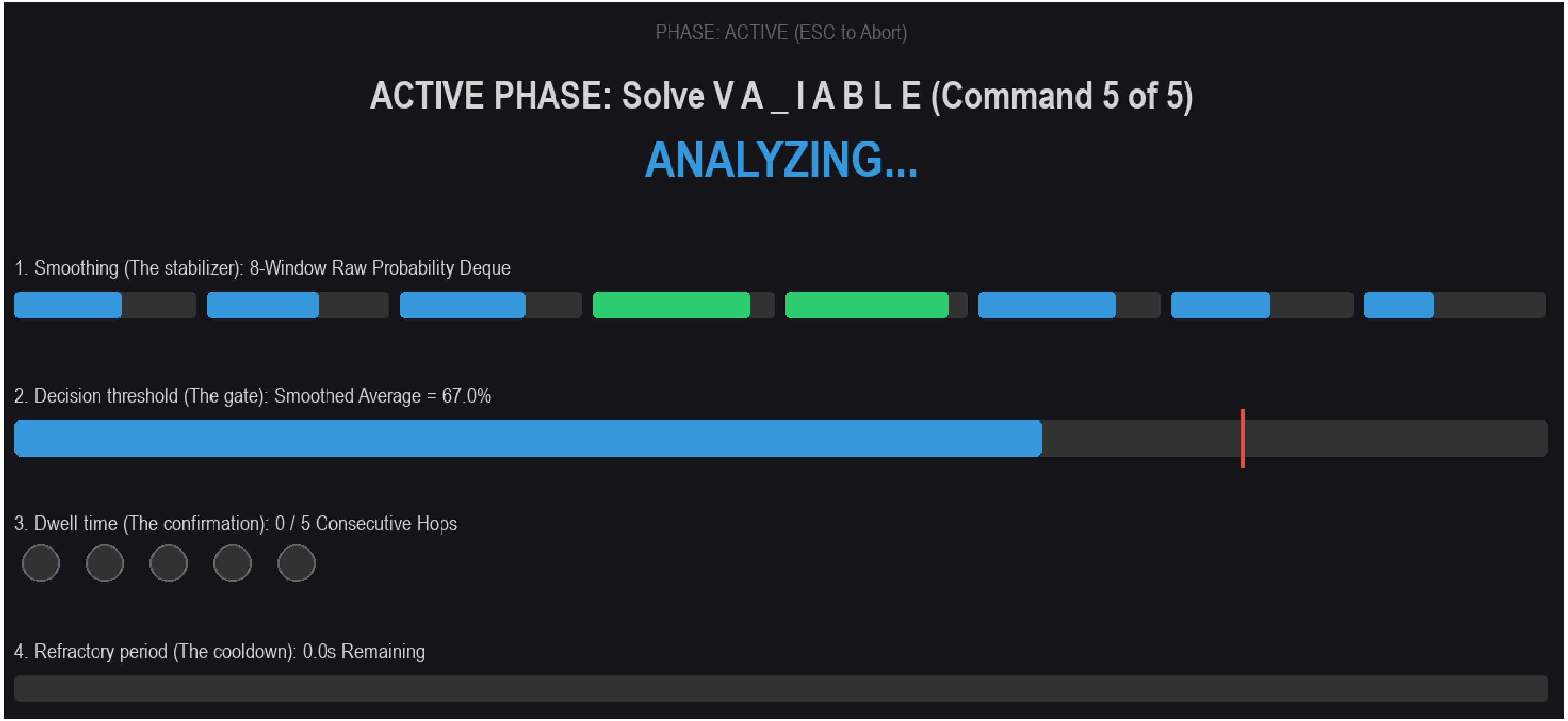
Dwell time (The confirmation)

- The smoothed score **must remain above 80%** for **5 consecutive hops** (1.25 seconds) to **confirm user intent** and reject random biological noise.
- Triggering a command requires **a minimum of 3.0 seconds of sustained focus**, combining **2.0** seconds to completely fill the initial smoothing buffer with **1.0** second (4 extra hops since the first hop out of 5 was achieved at the 2.0 seconds mark) to fulfill the remaining consecutive dwell time confirmation.

Refractory period (The cooldown)

- Upon triggering, a strict **3.0-second cooldown is enforced** where all new data is ignored, preventing rapid system overloads.

The decision engine



Automated evaluation and metrics

Task-level tracking

- Automatically tracks the **overall task completion rate**, in addition to the **average time required to trigger a command** and estimates the **system's ITR**.

System-level profiling

- Profiles the **end-to-end processing latency (median and worst-case)**, tracks **missed UI refresh deadlines** and calculates the **exact percentage of data windows refused due to artifacts**.

UX tracking

- **Logs false activations (commands misfired during the resting phase)** to measure system reliability under no-load conditions.

Automated export

- Instantly compiles all logged metrics into a **timestamped .txt report file** saved directly to the subject's online data folder upon completion.

Understanding the latency budget

The processing deadline (The analysis's budget)

- The strict computational time limit the CPU **has to finish all math before** the next chunk of live brain data arrives.
- Because our data window slides forward **every 250 ms** (a 4 Hz refresh rate), the **absolute processing budget is exactly 250 ms**.
- **Signal filtering, model inference, and visual UI rendering combined** only take about 29 ms. The system easily beats the deadline, preventing frame drops or background lag.

Total end-to-end delay (The user's experience)

- **The physical, perceived lag from the exact millisecond a neuron fires in the brain to the moment the UI physically changes on the screen.**
- Delay is cumulative. The system must account for **hardware transmission buffers (~25 ms), intentionally waiting to fill the next data window (250 ms), plus the actual computation (~29 ms)**.
- The user experiences a **total round-trip delay** of approximately 304 ms.

Automated metrics (numerical examples)

Task completion rate — 80.0%: Indicates the user successfully triggered 4 out of the 5 requested commands within the active phase time limits.

Avg time per command — 6.25 seconds: The **average total time from the prompt appearing to the final trigger**. This includes the initial mental effort to reach 80% confidence, plus the mandatory 1.25 seconds of sustained focus to lock the command in.

Estimated ITR — 8.50 bits/min: The ITR calculates the effective bandwidth of the BCI; a higher number indicates a faster, more accurate control loop.

End-to-end processing latency — 18.2ms (Median) / 45.1ms (Worst case): Proves the **DSP extraction and model inference** are executing well within the required real-time parameters (**processing much faster than the 250ms window hop**).

Missed deadlines — 0 loops exceeded 250 ms budget: Confirms the 4 Hz UI refresh rate remained perfectly stable without visual lag.

Artifact rejection — 12.5% (40/320 windows refused): Indicates that 12.5% of the live EEG data was thrown out due to the user blinking, clenching their jaw, or a sensor coming loose (exceeding the thresholds).

False activations — 1 (Commands fired during rest phase): Reveals the system misfired one time during the 30-second baseline resting phase, highlighting a potential false-positive flaw in the classifier or a failure by the user to properly relax.

Muse 2 online pipeline

Closing the loop

System architecture and utilized libraries

multiprocessing	Manages the concurrent execution of the hardware stream, operator terminal, and participant game.
os, sys, time	Handles system-level operations, precise latency tracking, and directory routing.
ctypes	Applies Windows OS-level overrides (HWND_TOPMOST) to lock focus on the Pygame windows.
numpy (np)	Powers high-performance matrix operations and buffer averaging.
scipy.signal	Applies causal DSP filtering (iirnotch, butter) and extracts Welch's PSD.
joblib	Dynamically deserializes the pre-trained scikit-learn ML models (LDA/SVM) and feature scalers.
muselsl and pylsl	Resolves the Bluetooth Low Energy (BLE) stream and manages the LSL inlet.
pygame	Drives the high-framerate, dual-monitor visual interfaces (operator and participant views).
tkinter	Renders the lightweight pre-registration GUI for session parameter selection.

Pipeline overview (The multi-process paradigm)

A synchronized three-process architecture

Process 1: Background hardware streamer

Establishes the BLE connection to the Muse headset and continuously broadcasts the raw EEG data via LSL.

Process 2: Operator diagnostics terminal

An independent window dedicated to monitoring signal integrity, system state, latency budgets, and real-time gamification progress (score and time remaining).

Process 3: Smart energy game engine

The **closed-loop participant interface** that processes ML inferences, enforces gamified time constraints, maps brain states to visual feedback, and logs final session outcomes.

Pipeline stage 1 – Pre-flight and acquisition

Task-specific loading

- The GUI automatically scans the /online/ directory and mounts the winning .joblib bundle based on the operator's task selection (Math vs. Missing Letter).

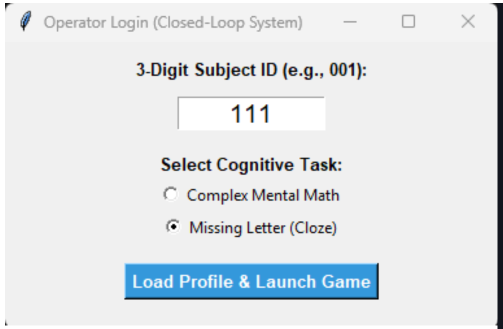
Stream resolution

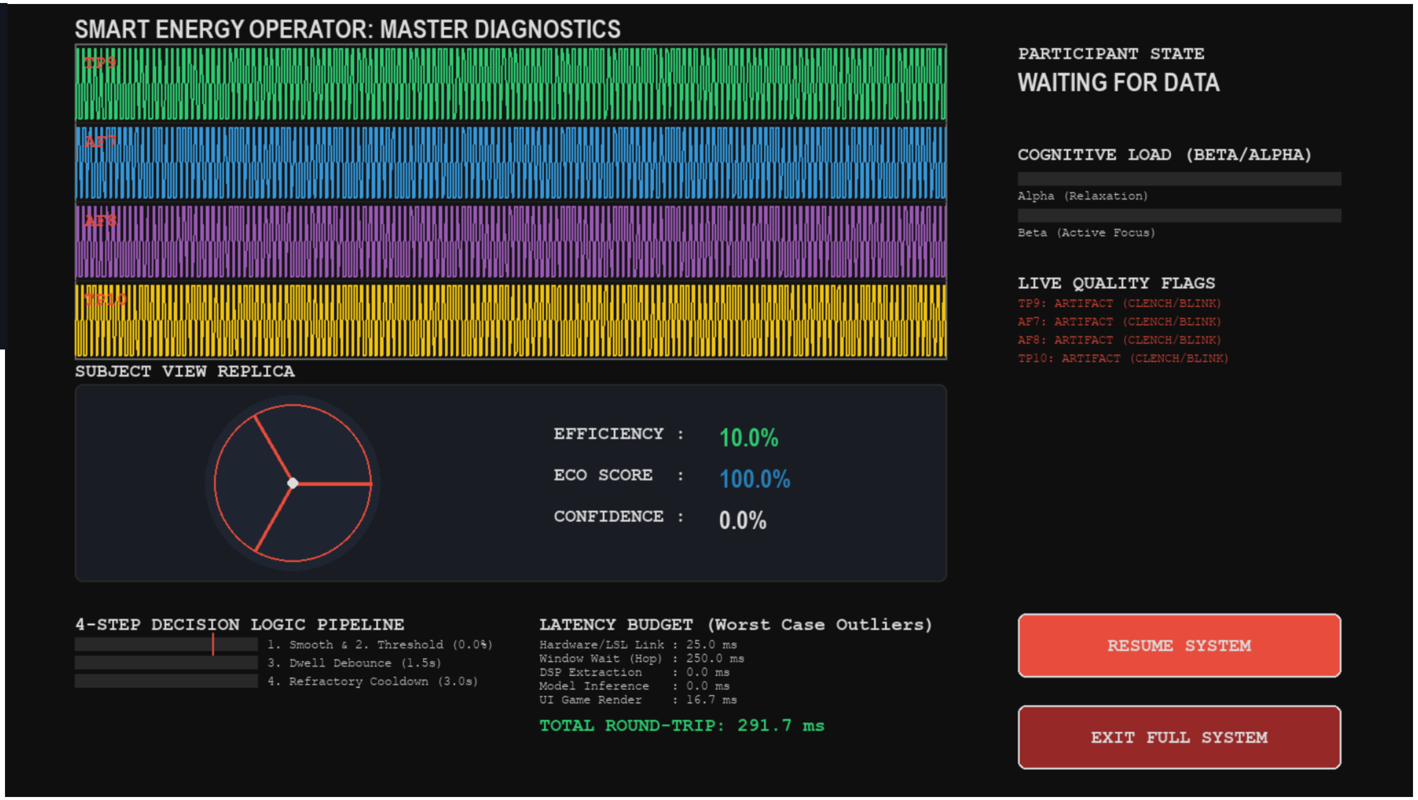
- The system enters a 25-second wait phase, actively scanning the local network for the type='EEG' LSL broadcast.

Epoch buffering

- Once connected, the engine initializes a rolling deque buffer, capturing a 2.0-second window of continuous data across TP9, AF7, AF8, and TP10.

Pipeline stage 1 – Pre-flight and acquisition





Pipeline stage 2 – Real-time DSP and inference

Sliding window hop

- The engine steps forward every 250ms (4 Hz refresh rate) to ensure snappy visual feedback without overwhelming the CPU.

Causal filtering

- Raw arrays pass through a 50 Hz notch filter and a 1–40 Hz bandpass filter to eliminate line noise and irrelevant biological frequencies.

Feature parity

- The engine extracts absolute power via Welch's method and calculates the exact Beta/Alpha ratio, mimicking the offline MNE pipeline.

Live prediction

- The extracted vector is normalized via the loaded standard scaler and fed into the prediction function to generate a raw confidence score.

Pipeline stage 3 – The operational decision engine

The stabilizer (Smoothing)

- The raw probability enters a **4-window rolling average** (1.0 second of history) to suppress erratic spikes.

The gate (Thresholding)

- The smoothed average must definitively cross the **strict 75% (0.75)** confidence boundary.

The confirmation (Dwell time)

- The signal must hold above 75% for **1.0 consecutive** seconds to confirm user intent and reject random biological noise.

The cooldown (Refractory)

- Upon triggering a command, a strict 2.0-second cooldown is enforced, ignoring all new data to prevent rapid, accidental double-firing.

The pipeline transitions from **highly conservative parameters** during the **calibration stage** to ensure pristine baseline data collection, to **faster, more relaxed thresholds** in the **operational stage** to prioritize a responsive, lag-free user experience.

Pipeline stage 4 – Operator diagnostics

Latency budget tracking

- Tracks millisecond-level delays across **DSP extraction, ML inference, and rendering** to ensure round-trip times remain safely below the 350 ms limit.
- Establishes 350 ms as the **absolute performance ceiling** comprising the **250 ms window hop, hardware transmission, and processing overhead** to prevent the system from **falling behind the live data stream**.

Automated quality flags

- Continuously monitors the peak-to-peak variance of all four electrodes, instantly flagging flatlines (loose sensors) or high-voltage artifacts (jaw clenches/blinks).

Live oscilloscope

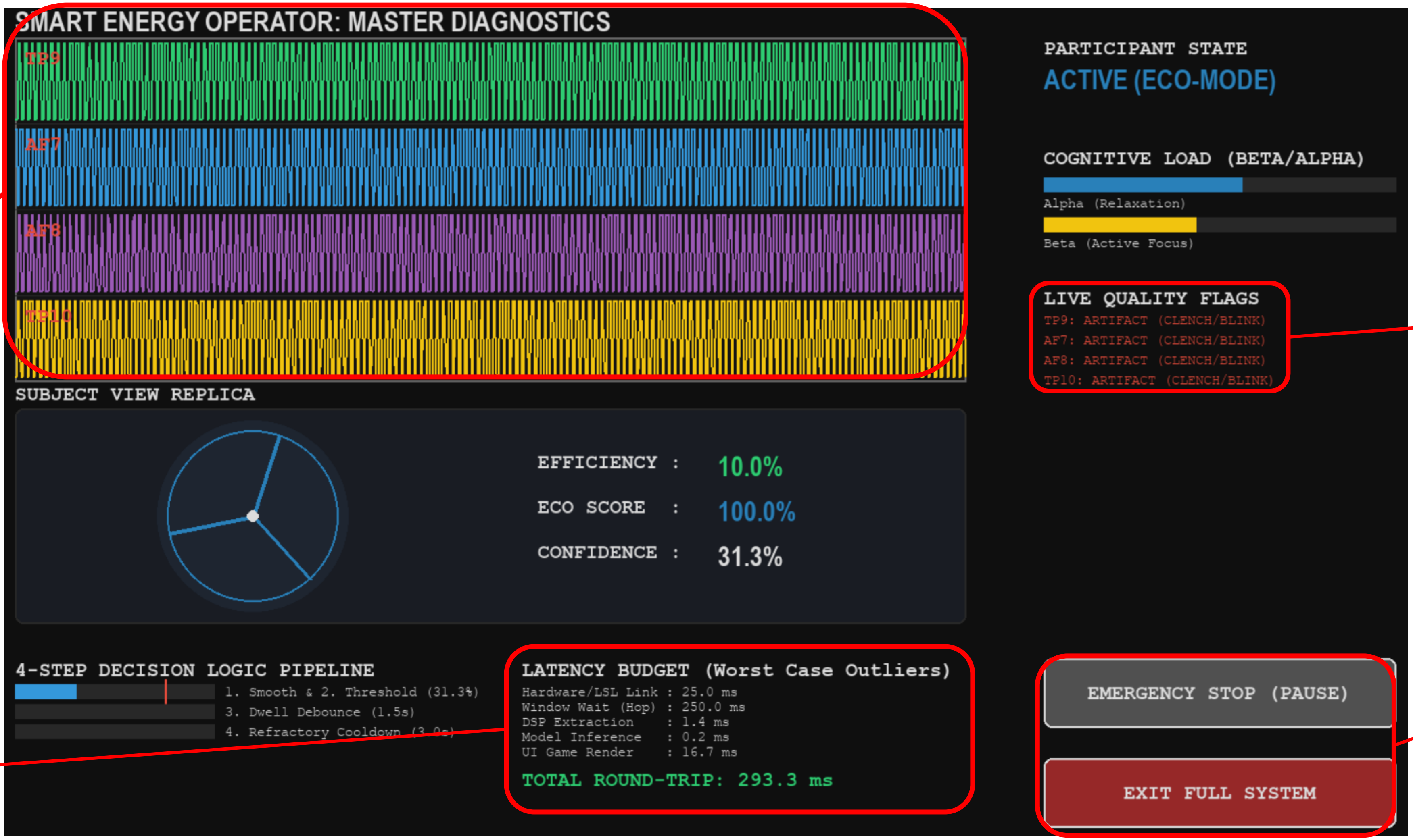
- Provides a real-time visual feed of the baseline-corrected EEG signals for visual inspection.

System overrides

- Features highly visible, mouse-clickable bounding boxes to instantly pause the system or trigger an emergency exit.

Pipeline stage 4 – Operator diagnostics

Live oscilloscope



Automated quality flags

System overrides

Latency budget tracking

Pipeline stage 5 – Closed-loop participant view (gamified)

Time-constrained gamification

- Participants are challenged to **achieve a target score of 10 successful commands within a strict 2-minute limit**, transitioning the paradigm from a free-run to an engaging, goal-oriented task.

Dynamic UI and telemetry

- The participant's dashboard is upgraded with a **live countdown timer (MM:SS)** and a **score progress tracker (X / 10)**, running alongside the **standard production efficiency** and **confidence metrics**.

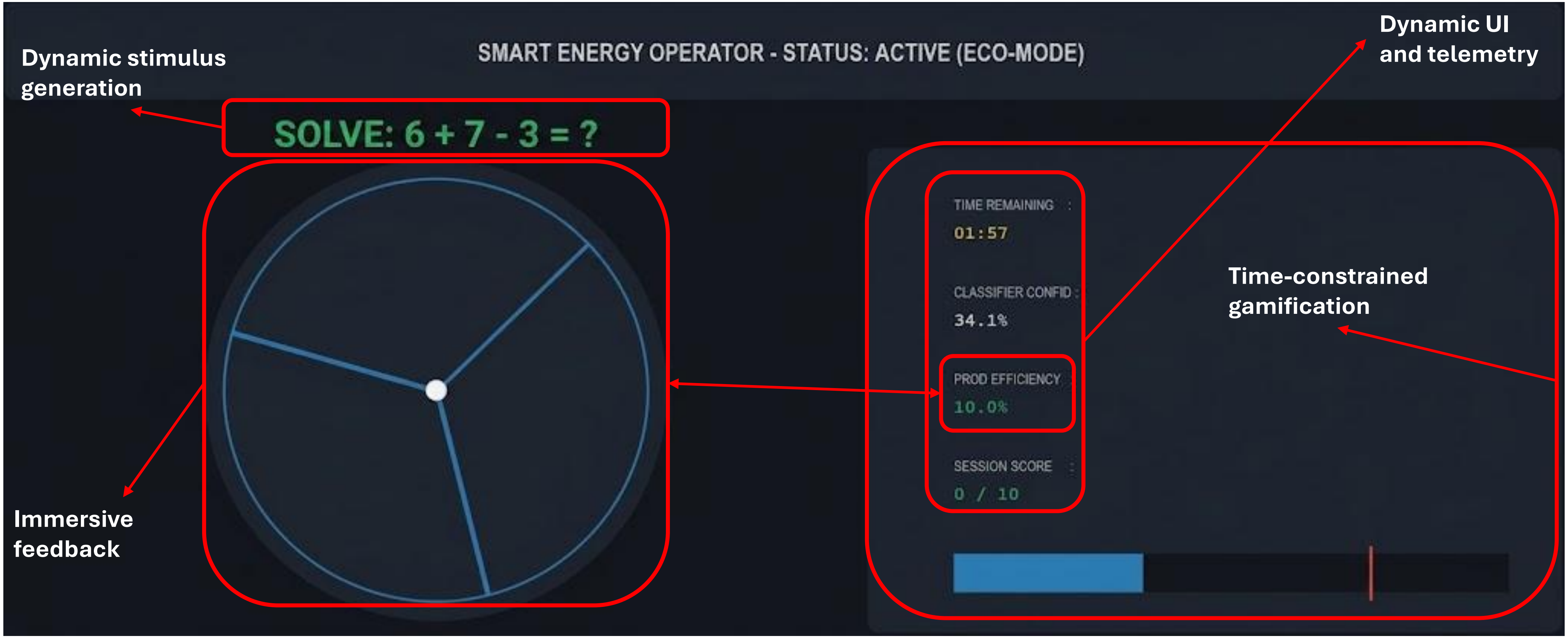
Dual end-state evaluation

- The session automatically halts upon reaching a definitive end-state: triggering **"MISSION ACCOMPLISHED!"** if the score is met early, or **"TIME EXPIRED"** if the 120 seconds elapse.

Automated audit logging

- Upon termination, the game engine seamlessly **appends the outcome status (SUCCESS vs. TIMEOUT), final score, and exact time taken to a dedicated, task-specific performance_log.txt file**.

Pipeline stage 5 – Closed-loop participant view



Interpreting the longitudinal performance log

The tracking mechanism	**What it logs**: The system autonomously records the participant's ID, specific cognitive task, final outcome (Success vs. Timeout), fractional score, and precise time to completion.
	Why it matters: Unlike isolated ML metrics, this log **measures functional BCI control**. It tracks the human's ability to reliably adapt their brain states (operant conditioning) under the pressure of a 2-minute countdown.
Diagnosing neurofeedback mastery	**The learning curve**: A **trend of consistently decreasing "Time taken" to reach the 10-point target** proves the user is transitioning from high-friction conscious effort to subconscious neuro-kinesthetic control.
	Replacing ITR: Instead of bits-per-minute, this closed-loop system **evaluates operator efficiency using a strict timeout paradigm**.
Real-world numerical example	**Result**: Score: 7/10 | Outcome: TIMEOUT | Time: 120.0s
	Interpretation: The participant experiences high cognitive friction. They are struggling to reliably push their Beta waves past the 75% threshold and sustain it for the required 1-second dwell time.

Technical advice and best practices

Respect the "Garbage in, garbage out" rule

- Always utilize the operator diagnostics terminal to **verify perfectly flat oscilloscope lines** before prompting the participant to begin the task.

Monitor the latency budget

- If your machine's CPU begins **thermal throttling, the "DSP extraction" latency will spike**.
- Keep an eye on the total round-trip time; **if it exceeds 350ms**, the system will begin missing frame deadlines.

Manage cognitive fatigue

- Ensure participants understand that the **2.0-second “refractory cooldown"** is their cue to breathe and soften their gaze.
- Sustaining extreme Beta-band focus for too long will result in **neural attenuation and system failure**.

Thank You!

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